MILITARY SPECIFICATION

THIONYL CHLORIDE
(For Use in Ammunition)

This specification approved for use by the US Army Armament Research and Development Command and is available for use by all Departmenst and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions and packaging for the manufacture of thionyl chloride for use in lithium ($SOCl_2$), reserve type batteries (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 Issue of Documents - The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipping and Storage
MIL-STD-1218 - ACS Chemicals

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

FSC: 6810

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASTM-E203	Water Using Karl Fischer Reagent
ASTM-E300	Sampling Industrial Chemicals
ASTM-D1217	Density and Specific Gravity of Liquids by Bingham Pycnometer
ASTM-D2905	Statements on Number of Specimens Required to Determine the Average Quality of a Textile Material

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19013).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and various Federal agencies).

3. REQUIREMENTS

3.1 Material. The material shall be thionyl chloride conforming to the following requirements: (see 6.3)

3.1.1 Appearance. The thionyl chloride shall be a clear, colorless to pale yellow volatile liquid with a very pungent odor when determined as specified in 4.6.1.

3.1.2 Properties. The thionyl chloride shall conform with the following table when determined as specified with the applicable paragraphs:

TABLE 1
Requirements for Thionyl Chloride

Properties	Requirements	Test Paragraph
Thionyl Chloride Percent (%) Minimum (Min)	99.6	4.6.2
Sulfur Dioxide % Maximum (Max)	0.3	4.6.2
Specific Gravity 25°/15.5°C	1.63	4.6.4
Iron, PPM. Max	20	4.6.3
Water, parts per million (PPM)	200	4.6.5
Sulfur Chloride % Max	0.1	4.6.2

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3.2 First Article Inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for Inspection and Standard Quality Assurance Provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspection. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3)
b. Quality Conformance Inspection (see 4.4)

4.3 First Article Inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2 The first article shall consist of 50 milliliters of thionyl chloride obtained by sampling as described in 4.5.1 (smaller size samples may be taken that are consistent with the requirements of the test). The sample shall be obtained from a production lot which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials, shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be Performed. The sample will be subjected by the Government to any or all of the examinations for tests specified in 4.6 of this specification.

4.3.3 Rejection. See MIL-A-48078.

4.4 Quality Conformance Inspection.

4.4.1 Lot Formation. A lot shall consist of one or more batches of thionyl chloride produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of thionyl chloride that has been subjected to the same unit chemical or physical

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process intended to make the final product homogeneous (See MIL-A-48078). The product shall be submitted for inspection in accordance with MIL-STD-105.

4.5 Testing (see 6.3) PRECAUTION. This specification covers sampling and testing of toxic or hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Sampling. Two (2) independent and representative samples (approx. 50 ml) of Thionyl Chloride shall be sampled as specified in ASTM E300. The samples shall be tested independently in accordance, with the applicable methods given in 4.6. If either sample fails to meet any of the test requirements, the lot shall be rejected.

The samples shall be subjected to the tests listed below:

Table II

Inspection/Requirement	Defect/Classification
Appearance (see 3.1.1)	Major
Properties (see 3.1.2)	Major

4.5.2 Inspection equipment The contractor shall use commercial inspection equipment for all tests and examination specified in par 4.6. He is also charged with the responsibility of assuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use.

4.6 Test Methods and Procedures. (see 6.2) - The tests to determine compliance with the requirements given in Table I shall be performed on the thionyl chloride samples obtained in 4.5.1. All tests given in this section shall be performed using prescribed procedures for replicate determinations given in standard analytical chemistry textbooks or ASTM D2905. Also, unless otherwise specified herein, all chemicals shall be ACS Grade in accordance with MIL-STD-1218 or reagent grade.

4.6.1 Appearance. The sample shall be examined visually to determine compliance with 3.1.1.

4.6.2 Thionyl Chloride, Sulfur Dioxide and Sulfur Chloride (see 6.2).

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4.6.2.1 Instrument conditions The thionyl chloride, sulfur dioxide and sulfur chloride shall be determined by gas chromatography using the following conditions.

Instrument: Perkin, Elmers, Model 900 or equivalent.

Detector: Thermal conductivity.

Columns: 2 meter glass tubing, 4mm diameter. Column packing: 10% QF-1 on Chromosorb G (AW. DMCS).

Oven temperature 53 - 85°C programmed 10°C per minute

Injection block temperature: 145°C.

Detector temperature: 200°C

Carrier gas: 50 ml/min. helium.

Sample size: 6 ul.

4.6.2.2 Procedure. Check the gas chromatographic operating conditions, make necessary adjustment and allow, the instrument to stabilize. Inject a 6 ul sample into the gas chromatograph set in accordance with the conditions specified in 4.6.2.1. Allow each component to elute from the column and measure its area with an integrator.

4.6.3 Determination of iron

4.6.3.1 Reagents and apparatus

Hydrochloric acid, conc.

Nitric acid, conc.

Iron standard solution, 10 Mg/ml

Pipettes, 20 ml, 10 ml, 5 ml

Beakers, 250 ml

Volumetric flasks, 50 ml

Atomic Absorption spectrophotometer - Varian Model AA5 or equivalent

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4.6.3.2 Procedure. Place 25 ml of concentracted hydrochloric acid into each of 2-250 ml beakers and then add 2 ml of concentrated nitric acid. Pipet 20 ml of sample (32.76 g) into each beaker. Allow vigorous action to subside and then place on a hot plate and evaporate to about 10 ml. Cool and dilute with distilled water. Transfer (filter if necessary into a 50 ml volumetric flask and make up to volume with distilled water. Adjust the instrument to the conditions given in 4.6.3.3.

With the digital readout control in the absorbance mode, aspirate the samples.

Measure and record the absorbance readings of each solution.

4.6.3.3 Parameters

	Iron
Lamp	Varian
Lamp current	25 ma
Slit width	50
Wavelength	248.3 nm
Oxidizer	air 7.5
Fuel	Acetylene 3.5

4.6.3.4 Calculations

$$\text{ppm Fe} = \frac{50\ A_1}{(A_2 - A_1)\ 32.76}$$

A_1 = absorption of solution without standard added

A_2 = absorption of solution with standard added

4.6.4 Specific Gravity. The specific gravity shall be determined in accordance with ASTM-D1217.

4.6.5 Determination of Water

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4.6.5.1 Infrared Spectrophotometer Method

4.6.5.1.1 Procedure. Working in an area having a relative humidity of 3 percent (glovebox or dry room) fill a dried 10mm cylindrical spectrophotometric cell having near-infrared (NIR) silica windows with thionyl chloride. This is best accomplished employing a glass pipette fitted with a rubber bulb. Stopper the cell with a ground glass or teflon stopper.

Place the filled cell in the sample beam of a Beckman Model 4240 or equivalent infrared spectrophotometer and identical unfilled cell in the reference beam. Scan from 4000 to 2500 cm^{-1} at a rate of 600 cm^{-1}/min.

Determine the percent transmission values (Po) and (P) for the peak at approximately 2800 cm^{-1}. Calculate the value of absorbance employing the following equation.

$$A = \log_{10} \frac{Po}{P}$$

If it is found that the absorbance of the 2800 cm^{-1} peak is greater than approximately 1.2, reanalyze the solution using a 5 mm cell.

4.6.5.1.2 Calculation

Calculate the water concentration of the thionyl chloride as follows:

1) 10mm Cell

$$ppm\ H_2O = \frac{A}{0.009033}$$

2) 5mm Cell

$$ppm\ H_2O = \frac{2A}{0.009033}$$

4.6.5.2 Alternate Method by Karl Fischer. The water shall be determined in accordance with ASTM-E203.

5. PACKAGING

5.1 Packing

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5.1.1 Level C. Unless otherwise specified the thionyl chloride shall be packed in accordance with manufacturer's commercial practice to assure acceptance by common carrier for safe delivery at first destination for immediate use.

5.2 Marking. Unless otherwise specified, the shipping containers shall be marked in accordance with MIL-STD-129, and shall include the lot number.

6. NOTES

6.1 Ordering data. See MIL-A-48078

6.2 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted thru the Contracting Officer to: Commander, ARRADCOM, ATTN: DRDAR-QAR-R, Dover, NJ 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.3 Thionyl chloride which has been found satisfactory for use in lithium batteries is manufactured by MCB. Manufacturing Chemist, Cincinnati, Ohio, 45212, Eastman Kodak Co., Rochester, N.Y. 14650, and Mobay Chemical Corporation, Penn Lincoln Parkway West, Pittsburgh, PA 15205. Thionyl chloride from other sources should be evaluated by ARRADCOM, ATTN: DRDAR-QAR-R, Dover, NJ 07801 before being approved as an equivalent.

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MIL-T-63466 (AR) THIONYL CHLORIDE (FOR USE IN AMMUNITION)

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